Can You Trust the Explanation? Auditing the Agreement and Stability of SHAP and LIME for Cryptocurrency Direction Models Across Volatility Regimes

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Abstract

Applied cryptocurrency-forecasting papers often present SHAP or LIME feature importances as facts about the market. We test whether such explanations can be trusted. XGBoost, LightGBM and a small MLP predict next-day direction for BTC, ETH, ADA, LINK and LTC under walk-forward validation (55 folds, 9835 test days per model). On 550 explained test days we measure cross-method agreement, stability under retraining, stability under small input perturbations, and whether instability differs between high- and low-volatility regimes. Every number is compared with the explainer's own seed-to-seed noise floor and with a shuffled-label control model that has no skill. The models barely beat a coin flip (accuracy 0.507–0.512 vs 0.502 for the control), with a small edge only in volatile regimes. Explanations are fragile. LIME shares about a third of its top-3 features with SHAP and 0.638 with itself. TreeSHAP, which has zero explainer noise, keeps a top-3 Jaccard of only 0.624 across retraining seeds, and a 5% input nudge changes its top feature for 0.280 of points. The no-skill control is equally stable. On the key question we find a clear null: no explanation-instability measure is significantly worse in volatile regimes (0 of 52 tests after Benjamini–Hochberg correction).

1 Introduction

Machine-learning models for cryptocurrency prices are now common, and many papers pair them with post-hoc explainers such as SHAP [1] or LIME [4] to report which indicators "drive" the predictions [7]. Such a report carries an implicit claim: that the explanation describes something stable about the model, and through the model, about the market. Both links are weak. Daily crypto direction is close to unpredictable, so a model may have little real structure to explain. Explanation methods are known to be sensitive to random seeds, sampling, correlated features and small input changes [2, 3, 5, 8], and different methods often disagree with each other [10].

This project asks how far the feature-importance story can be trusted in this setting, and under what conditions it breaks down:

- **RQ-a (agreement).** Do TreeSHAP, KernelSHAP, LIME and permutation importance agree on the top drivers?
- **RQ-b (retraining).** Are the drivers stable when the model is retrained with new seeds or on block-bootstrap resamples?
- **RQ-c (perturbation).** Are they stable when the inputs are perturbed slightly?
- **RQ-d (regimes, the key question).** Is the instability worse in high-volatility regimes than in calm ones?

Contributions.

1. A leakage-free audit pipeline: causal features (tested by truncation), expanding walk-forward validation, and a regime label computed only from training data.
2. Four audits run on the *same* 550 explained points, so agreement, retraining and perturbation results are directly comparable.
3. Two explicit baselines for every instability number: the explainer's own **noise floor** (common random numbers make the comparison exact) and a **shuffled-label control model**.
4. A careful regime comparison: cluster-bootstrap CIs, within-fold permutation tests, a placebo relabelling test, and Benjamini–Hochberg correction over 56 tests.
5. A clear answer: explanations of these models are fragile, the fragility is the same for a no-skill model, and it is *not* worse in volatile regimes.

2 Related Work

Post-hoc attribution methods. Ribeiro et al. [4] introduced LIME, which fits a weighted linear surrogate to random perturbations around a point. Lundberg and Lee [1] unified attribution methods under Shapley values and proposed KernelSHAP, a sampling-based estimator. TreeSHAP [11] computes Shapley values exactly for tree ensembles. Permutation importance [16, 9] measures the global drop in performance when one feature is shuffled. Molnar [9] surveys these methods and their caveats, especially with correlated features.

Instability and fragility. Alvarez-Melis and Jaakkola [2] showed that LIME and SHAP explanations can change sharply under small input changes. Ghorbani et al. [8] showed that explanations of neural networks are fragile under imperceptible perturbations. Slack et al. [3] built models that deliberately fool LIME and SHAP. Zhou, Hooker and Wang [5] traced LIME's instability to its random sampling and proposed S-LIME, which chooses the sample size by hypothesis testing so that explanations stabilise.

The disagreement problem. Krishna et al. [10] showed that practitioners routinely see explanation methods disagree on the top features of the same prediction. They defined top-k feature agreement, rank agreement and related metrics. Our top-3 Jaccard and Spearman metrics follow that framing.

ML for crypto prediction. Gradient-boosted trees [6, 15] and small neural networks are standard tools in applied crypto forecasting. Pinky and Akula [7] is a representative example of work that reports model accuracy together with feature-importance analysis. We found no study that measures agreement, retraining stability and perturbation stability *together*, against a noise floor and a no-skill control, split by volatility regime, for crypto direction models.

3 Data and Features

Data. Daily Coin Metrics Community Data [14] for BTC, ETH, ADA, LINK and LTC: `PriceUSD` (a reference price at 00:00 UTC) and `volume_reported_spot_usd_1d`, from 2017-12-01 to 2026-05-22. The loader refuses gaps and duplicates rather than filling them. The data are close-only (no open/high/low). LTC replaces the planned SOL because SOL's volume history is missing in this dataset.

Features. Twelve features use information up to the end of day t only: log returns at lags 1–3; 5- and 10-day cumulative returns; Wilder RSI(14); MACD, its signal line and histogram (each divided by price); 7- and 30-day realised volatility; and log volume relative to its 30-day mean. The MACD trio

is exactly collinear (`macd_hist = macd − macd_signal`). We keep it on purpose, because correlated features are common in practice and make attribution methods split credit. The **target** is 1 if the next day's log return is positive. After 60 warm-up rows, each coin has 3034 rows. Formulas are in `docs/methods_and_metrics.md`. Causality is tested by rebuilding the features on truncated series and by injecting a future price shock (tests TC-01 to TC-04).

4 Methodology

Walk-forward validation. Expanding window. 182-day test blocks start on 2021-01-01. Each fold trains on every row dated before the block, minus a 1-day embargo, and needs at least 730 training rows. Nothing is shuffled (TC-05).

Volatility regime. A test day is *volatile* if its trailing 30-day volatility (`vol_30`) exceeds the median `vol_30` of the fold's **training** window; otherwise it is *calm*. The threshold uses training data only, so the label is known on the prediction day (TC-06, TC-07).

Models. XGBoost (200 trees, depth 3) and LightGBM (200 trees, 8 leaves), both with 0.8 row and column subsampling, and an MLP (32–16 hidden units, early stopping, standardised inputs). The **control** `xgb_shuffled` is the same XGBoost trained on randomly permuted training labels, so it has no skill by construction.

Explainers. *TreeSHAP* (path-dependent, log-odds; tree models only). *KernelSHAP* (logit link, 20-centre k-means background, 200 coalition samples; the shap default `l1_reg = num_features(10)` keeps 10 of 12 features non-zero). *LIME* (quartile discretisation, 1000 samples, probability space). *Permutation importance* (drop in ROC-AUC on the test fold, 10 repeats; global only). We compare rankings of |attribution|, never raw values, because the output spaces differ.

Metrics. Our primary metric is **top-3 Jaccard**: the overlap of the two sets of three features with the largest |attribution| (1 = same set). We also use Spearman correlation of |attr| over all 12 features, **top-1 agreement**, the **top-1 change rate** (how often the top feature differs from a reference explanation), and the **prediction flip rate** (how often the predicted class changes). As calibration, two *random* rankings of 12 features have expected top-3 Jaccard 0.164 and top-1 agreement 0.083.

Audits. (a) *Agreement*: every pair of explainers on the same point, and globally permutation importance vs mean |attribution| per fold. (b) *Noise floor*: KernelSHAP and LIME re-run with new seeds on the identical model and input; TreeSHAP is re-run and asserted bit-identical. (c) *Retraining*: refits with 5 new seeds and on 5 moving-block bootstraps (30-day blocks) of the training rows [13], each re-explained with the model's primary method (TreeSHAP for trees, KernelSHAP for the MLP). (d) *Perturbation*: Gaussian noise of $\varepsilon \times$ (training std) for $\varepsilon \in \{0.01, 0.05\}$, with `macd_hist` recomputed and volatilities kept ≥ 0. Perturbation repeat r reuses the explainer seed of noise-floor repeat r, so the gap between the two is caused only by the input change.

Statistics. Points within a (coin, fold) share a model and a market period, so we treat the 55 (coin, fold) pairs as **clusters**. CIs are 95% **cluster bootstraps**. Each volatile − calm difference is tested with a **within-cluster permutation test**. Every explanation metric is oriented as *instability* (1 − Jaccard, or the top-1 change rate), so a positive difference means "less stable when volatile". **Benjamini–Hochberg** [12] controls the false discovery rate over all 56 tests. In the **placebo test**, each fold receives the regime sequence of a *different*, randomly chosen fold, which gives realistic

but meaningless labels. The resulting null distribution shows how large a regime gap can appear by chance.

5 Experimental Setup

Table 1: Experimental setup. Sources: `results/run_log.json`, `results/tables/folds.csv`, `key_numbers.json`.

Item	Value
Coins / rows per coin	5 / 3034
Walk-forward folds	11 per coin, 55 total; training 1066–2886 rows; test blocks 147–182 days
Out-of-sample days per model	9835 (1967 per coin); 0.293 of them volatile
Models	xgb, lgbm, mlp + xgb_shuffled control, all fit at `base_seed = 0`
Explained points	10 per (coin, fold) → 550 (310 calm, 240 volatile); 7 of the 55 folds had no volatile test day
Noise floor	3 extra seeds per stochastic explainer
Retraining	5 seeds + 5 moving-block bootstraps (block 30) per model and fold
Perturbation	$\varepsilon \in \{0.01, 0.05\}$; 10 repeats (TreeSHAP), 3 repeats (KernelSHAP, LIME)
Statistics	2000 cluster-bootstrap resamples, 2000 permutations, 2000 placebo relabellings; 56 tests
Software	Python 3.12.15, numpy 2.5.4, pandas 3.0.6, scikit-learn 1.9.1, xgboost 3.4.1, lightgbm 4.7.0, shap 0.54.1, lime 0.2.0.1
Runtime	310.3 s for the full run on an Apple-silicon laptop with 2 worker processes

All randomness derives from `base_seed` through CRC-hashed task keys. Two runs give identical CSVs (test TC-20), and a run from a fresh clone reproduced every committed result file byte-for-byte. The full run was well under the 60-minute budget, so all sample sizes are the configured defaults. All 25 tests pass.

6 Results

Every stability table below lists the model's out-of-sample accuracy next to it: xgb 0.512, lgbm 0.512, mlp 0.507, control 0.502.

6.1 Predictive performance

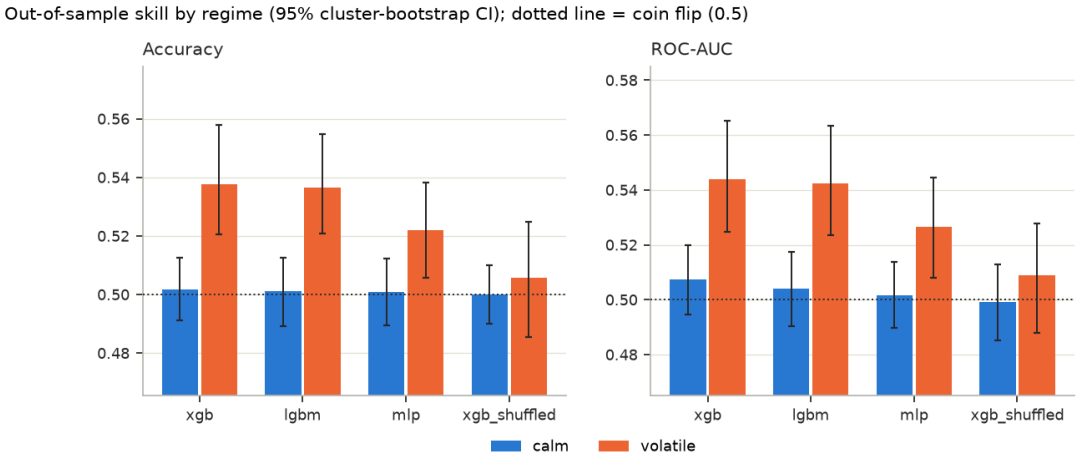


Figure 1: Out-of-sample accuracy and ROC-AUC by regime, with 95% cluster-bootstrap CIs. The dotted line is a coin flip.

Table 2: Accuracy over 9835 test days per model. "Always predict up" scores 0.504 overall (0.506 calm, 0.496 volatile). Sources: `accuracy_by_regime.csv`, `regime_tests.csv`.

model	accuracy	95% CI	AUC	calm acc.	volatile acc.	volatile − calm (q)
xgb	0.512	0.502–0.523	0.520	0.502	0.538	+0.036 (0.028)
lgbm	0.512	0.501–0.522	0.517	0.501	0.537	+0.035 (0.028)
mlp	0.507	0.498–0.517	0.509	0.501	0.522	+0.021 (0.150)
xgb_shuffled (control)	0.502	0.493–0.511	0.502	0.500	0.506	+0.006 (0.758)

The models are barely better than a coin flip. The MLP's CI includes 0.5, and the tree models' CIs start at 0.502 and 0.501. In **calm** regimes no model beats the control: all score 0.500–0.502. In **volatile** regimes XGBoost and LightGBM reach 0.538 and 0.537 (AUC 0.544 and 0.542), ahead of the control (0.506) and "always up" (0.496). This difference survives BH correction, and it lies outside the placebo 95% range (xgb: −0.023 to 0.021). It is not universal: for LINK, XGBoost scores 0.498 overall and 0.492 in volatile regimes (`accuracy_by_coin.csv`). **Every explanation below therefore describes a model that is mostly fitting noise.** That is typical for next-day crypto models, and it is exactly when feature-importance stories are most tempting and least justified.

6.2 Cross-method agreement

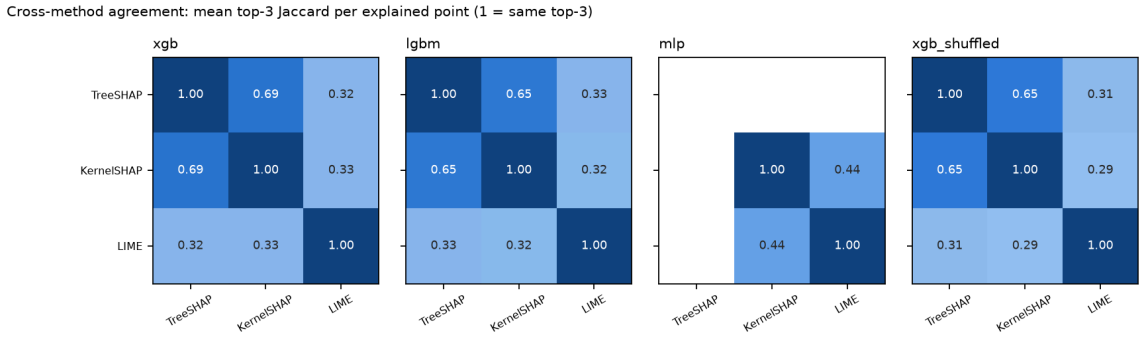


Figure 2: Mean per-point top-3 Jaccard between explainers (1 = same top-3). Random rankings would score 0.164.

Table 3: Per-point top-3 Jaccard, with top-1 agreement in brackets (550 points per cell). Last column: LIME's top-3 Jaccard against itself across seeds. Sources: `agreement_local_summary.csv`, `noise_floor_summary.csv`.

model (acc.)	TreeSHAP–KernelSHAP	TreeSHAP–LIME	KernelSHAP–LIME	LIME vs itself (noise floor)
xgb (0.512)	0.686 (0.742)	0.324 (0.331)	0.329 (0.355)	0.638
lgbm (0.512)	0.653 (0.747)	0.328 (0.335)	0.319 (0.324)	0.631
mlp (0.507)	–	–	0.438 (0.453)	0.678
xgb_shuffled (0.502)	0.651 (0.740)	0.314 (0.276)	0.288 (0.269)	0.619

TreeSHAP and KernelSHAP share about two-thirds of their top-3 features. Both are Shapley-value methods, but path-dependent TreeSHAP and interventional KernelSHAP treat correlated features differently. LIME agrees with either SHAP variant at about 0.32: above the random baseline (0.164), but far below SHAP–SHAP. **Part of that gap is LIME's own noise**: run on the identical model and point with a new seed, LIME agrees with itself at only 0.619–0.678. This is the instability S-LIME [5] was designed to fix.

Globally, permutation importance agrees poorly with mean |TreeSHAP|: per-fold top-3 Jaccard 0.267 (xgb) and 0.278 (lgbm), Spearman 0.286 and 0.279. For the MLP, Spearman with mean |KernelSHAP| is 0.101 (`agreement_global_summary.csv`). This is what near-zero skill predicts. Permutation importance measures a feature's effect on *accuracy*, which is close to zero for every feature, while SHAP measures its effect on the *output*.

Control comparison. The no-skill control agrees about as much as the real models (TreeSHAP–KernelSHAP 0.651 vs 0.686 for xgb). Agreement between explainers is not evidence that the model learned anything.

6.3 Retraining stability

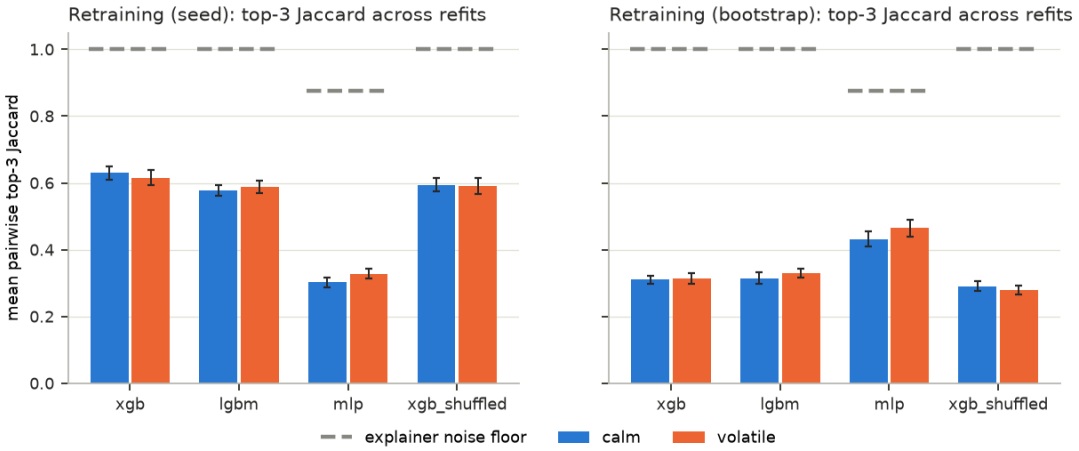


Figure 3: Retraining stability: top-3 Jaccard across the base model and 5 refits, by regime. Dashed line: the primary explainer's noise floor (1.0 for TreeSHAP).

Table 4: Mean pairwise stability among the base model and 5 refits (550 points each). Noise floor = the same explainer's top-3 Jaccard against itself on an unchanged model. Class flips = share of refits whose predicted class differs from the base model's. Source: `retrain_summary.csv`.

model (acc.), method	seeds: Jaccard [95% CI]	seeds: top-1 same	bootstraps: Jaccard [95% CI]	noise floor	class flips (seeds / bootstraps)
xgb (0.512), TreeSHAP	0.624 [0.607, 0.641]	0.664	0.312 [0.301, 0.323]	1.000	0.128 / 0.280
lgbm (0.512), TreeSHAP	0.582 [0.569, 0.595]	0.662	0.320 [0.309, 0.332]	1.000	0.150 / 0.279
mlp (0.507), KernelSHAP	0.313 [0.302, 0.325]	0.284	0.446 [0.425, 0.467]	0.876	0.412 / 0.280
xgb_shuffled (0.502), TreeSHAP	0.593 [0.577, 0.609]	0.633	0.285 [0.274, 0.297]	1.000	0.140 / 0.288

Changing only XGBoost's random seed leaves a top-3 Jaccard of 0.624. The top-1 feature agrees in only 0.664 of model pairs, and across 6 models a point has on average 1.936 distinct top-1 features. TreeSHAP's noise floor is exactly 1, so **all of this is model sensitivity, not explainer noise**. Block-bootstrap resampling is more disruptive still: Jaccard drops to 0.312, not far above the random baseline of 0.164, and the predicted class flips in 0.280 of refits. For the MLP, the KernelSHAP noise floor is 0.876, but retraining stability is 0.313 (seeds) and 0.446 (bootstraps), far below it. Accuracy barely moves across refits: XGBoost's pooled accuracy across seeds ranges from 0.509 to 0.516.

Control comparison. The control is nearly as stable as the real models (0.593 vs 0.624 across seeds). There is one sign of learning, at the *global* level. Across folds, the real XGBoost puts `ret_lag1` in its TreeSHAP top-3 in 0.927 of folds, while the control's most frequent top-3 feature (`ret_5d`) reaches only 0.418 (`global_feature_ranks.csv`; see Figure 4).

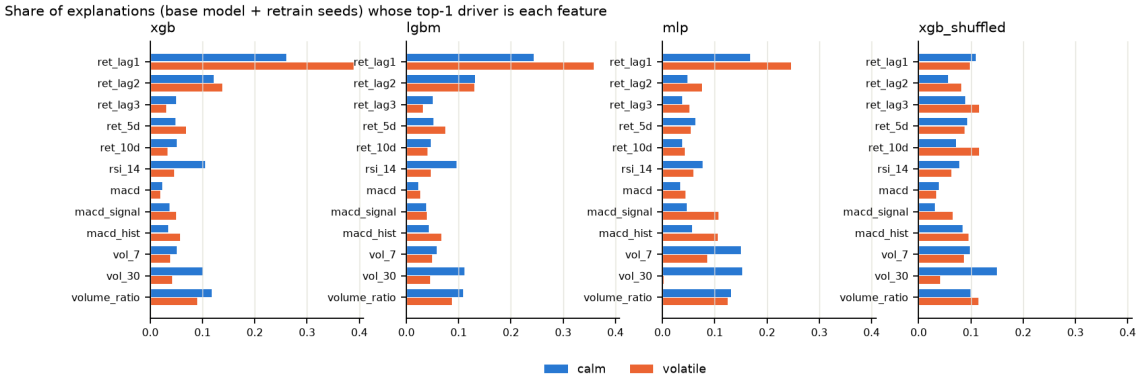


Figure 4: Share of explanations (base model + 5 seeds) whose top-1 feature is each feature, by regime.

6.4 Perturbation stability

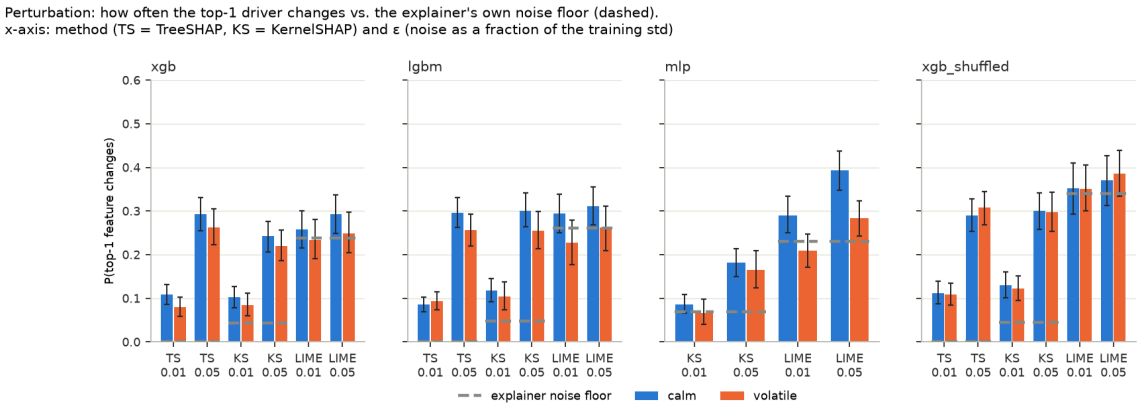


Figure 5: P(top-1 feature changes) under ε × std Gaussian input noise, by regime. Dashed line: the explainer's noise floor (top-1 change rate between two seeds on the unperturbed input).

Table 5: P(top-1 feature changes) relative to the unperturbed explanation (550 points each). The excess over the noise floor is paired per point. Sources: `perturbation_summary.csv`, `perturbation_excess_over_noise_floor.csv`.

model (acc.)	method	ε = 0.01	ε = 0.05	noise floor	excess at ε = 0.05 [95% CI]	prediction flips (ε = 0.05)
xgb (0.512)	TreeSHAP	0.096	0.280	0.000	0.280	0.136
xgb (0.512)	KernelSHAP	0.095	0.232	0.043	0.189 [0.165, 0.214]	0.133
xgb (0.512)	LIME	0.247	0.273	0.238	0.036 [0.021, 0.052]	0.133
lgbm (0.512)	TreeSHAP	0.089	0.278	0.000	0.278	0.142
mlp (0.507)	KernelSHAP	0.078	0.175	0.068	0.106 [0.081, 0.130]	0.059
mlp (0.507)	LIME	0.255	0.345	0.231	0.114 [0.091, 0.136]	0.059
xgb_shuffled (0.502)	TreeSHAP	0.110	0.298	0.000	0.298	0.145
xgb_shuffled (0.502)	KernelSHAP	0.127	0.299	0.044	0.255 [0.222, 0.290]	0.148
xgb_shuffled (0.502)	LIME	0.351	0.377	0.339	0.038 [0.018, 0.056]	0.148

Gaussian noise of 5% of each feature's training standard deviation changes the TreeSHAP top-1 feature for 0.280 (xgb) and 0.278 (lgbm) of points. Even 1% changes it for 0.096 and 0.089. For the tree models, the explanation changes about twice as often as the predicted class does (0.136 at ε = 0.05). Measured against the **noise floor**: for KernelSHAP the floor is low and the excess is real (0.189 for xgb at ε = 0.05). For LIME, most of the change is noise. At ε = 0.01 its excess is 0.010 (xgb) and 0.005 [−0.002, 0.013] (lgbm), so a LIME top-1 change at that size is almost entirely sampling noise. **Control comparison:** the control is as fragile as the real models (TreeSHAP 0.298 at ε = 0.05).

6.5 Volatility-regime comparison (the key question)

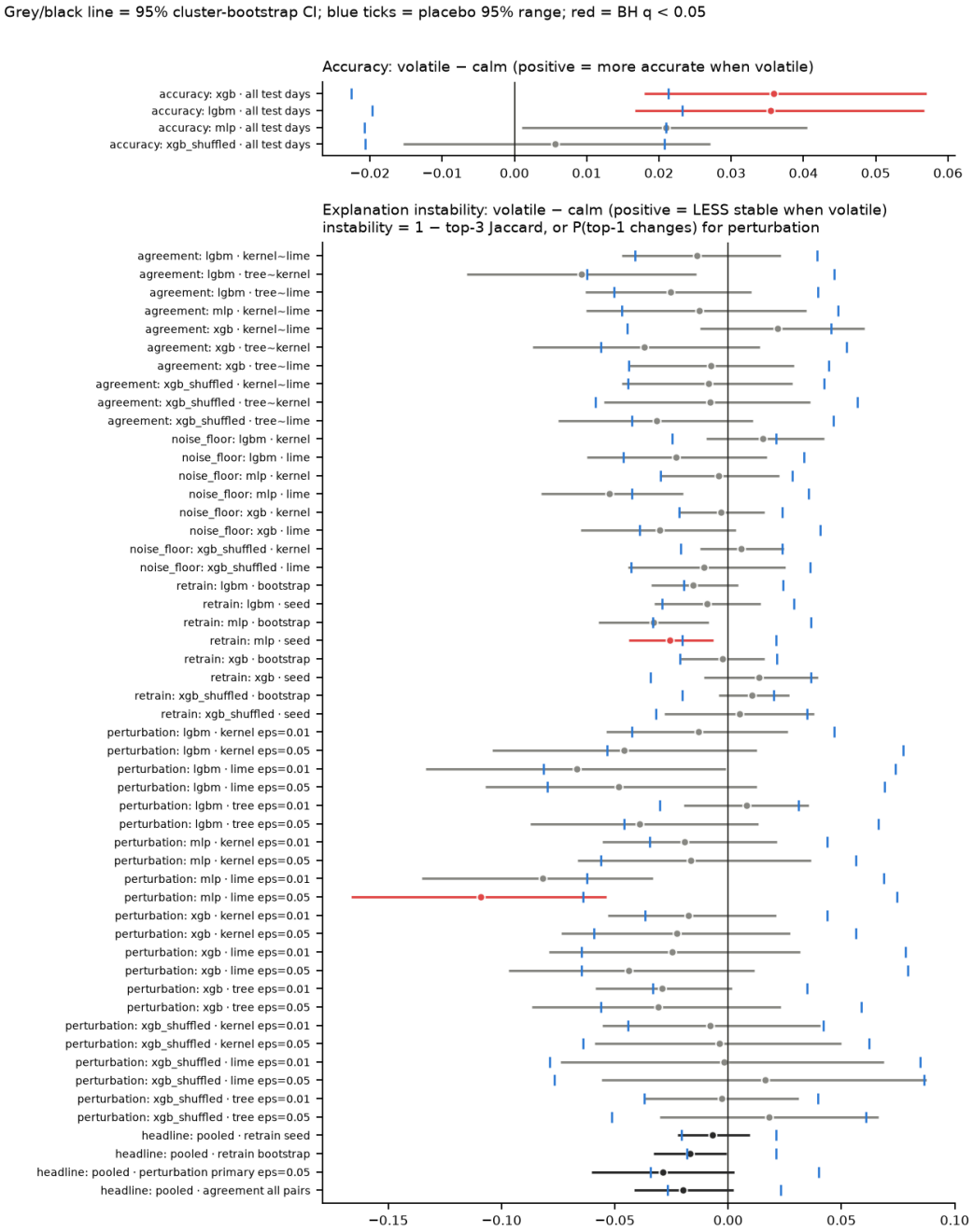


Figure 6: Volatile − calm differences for all 56 tests. Lines: 95% cluster-bootstrap CI. Blue ticks: placebo 95% range. Red: BH q < 0.05. Lower panel: positive = less stable when volatile.

Table 6: Pooled headline regime tests over xgb, lgbm and mlp. Instability = 1 − top-3 Jaccard (retraining, agreement) or P(top-1 changes) (perturbation, primary explainer). Rows: 930 calm / 720 volatile point-model pairs; agreement 2170 / 1680 point-pairs. Source: `regime_tests.csv`.

test (pooled models)	calm	volatile	volatile − calm [95% CI]	p (perm.)	q (BH)	placebo 95% range	p (placebo)
retrain, seeds	0.497	0.490	−0.007 [−0.022, 0.009]	0.492	0.758	−0.021 to 0.021	0.507
retrain, bootstraps	0.648	0.631	−0.017 [−0.033, −0.001]	0.064	0.258	−0.018 to 0.021	0.093
perturb., ε = 0.05	0.257	0.228	−0.029 [−0.060, 0.003]	0.029	0.150	−0.034 to 0.040	0.133
agreement	0.569	0.549	−0.020 [−0.041, 0.002]	0.102	0.336	−0.027 to 0.023	0.130

Of the 56 tests, 12 have a raw permutation $p < 0.05$, and 4 survive BH correction at $q < 0.05$ (`key_numbers.json`):

- accuracy, xgb: +0.036 (q = 0.028);
- accuracy, lgbm: +0.035 (q = 0.028);
- retraining (seeds), mlp: 1 − Jaccard −0.026 (q = 0.028), i.e. **more** stable when volatile;
- perturbation, mlp/LIME, ε = 0.05: top-1 change rate −0.109 (q = 0.028), i.e. **more** stable when volatile.

No explanation-instability measure is significantly higher in volatile regimes. Only 9 of the 52 explanation differences are positive at all. All four pooled headline differences are slightly negative and fall inside their placebo 95% ranges. The two significant explanation differences (both MLP) point *against* the hypothesis. The control's regime differences are all non-significant (smallest q = 0.570). The only robust regime effect is in **accuracy**, not in explanation stability.

6.6 Noise floor and placebo

Table 7: Explainer noise floors: the same model and input, re-explained with 3 new seeds (550 points each). Source: `noise_floor_summary.csv`.

model	KernelSHAP vs itself: Jaccard	KernelSHAP: top-1 change rate	LIME vs itself: Jaccard	LIME: top-1 change rate	TreeSHAP
xgb	0.938	0.043	0.638	0.238	1.000 / 0.000
lgbm	0.922	0.047	0.631	0.261	1.000 / 0.000
mlp	0.876	0.068	0.678	0.231	–
xgb_shuffled	0.945	0.044	0.619	0.339	1.000 / 0.000

Noise floor. TreeSHAP is exactly reproducible. KernelSHAP is close (top-3 Jaccard 0.876–0.945 against itself). LIME is noisy: its top-1 feature changes in 0.231–0.339 of seed-only reruns. The noise floors do **not** differ by regime: none of the 8 noise-floor regime tests is significant (smallest q = 0.150). So the regime comparison is not confounded by explainer noise.

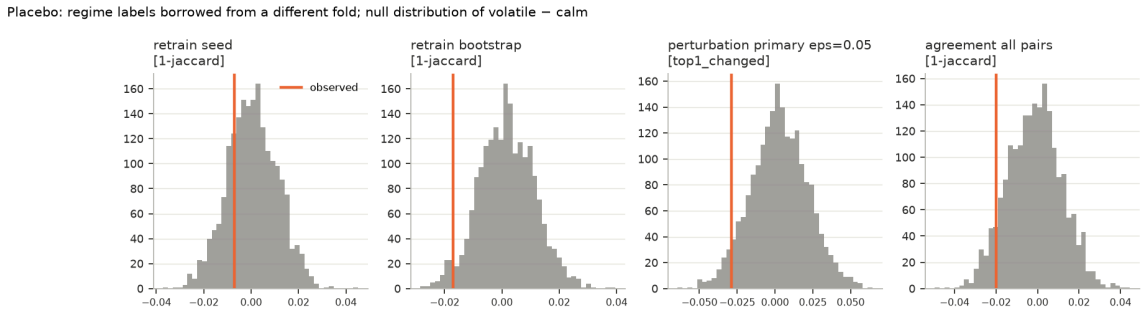


Figure 7: Placebo null distributions (regime labels borrowed from a different fold, 2000 times) for the four headline tests, with the observed difference.

Placebo. Random but realistic regime labels produce volatile – calm gaps whose 95% ranges span from −0.034 to 0.040 on the headline measures (Table 6). Every observed headline explanation difference falls inside its placebo range. In contrast, the accuracy gaps (+0.036 and +0.035) lie outside theirs (xgb placebo range −0.023 to 0.021, placebo p = 0.0005), which confirms that the test can detect a real regime effect when one exists.

7 Discussion

What the results mean. A per-prediction "top driver" from these models depends on the explainer (6.2), the random seed and training sample (6.3), and tiny input changes (6.4). A model with no skill gives explanations that are just as consistent, so consistency cannot be used to argue that an explanation reflects the market. The one more robust signal is global: across folds the real tree models keep ranking the most recent return (`ret_lag1`) highly, which the control does not.

Why is there no regime effect? We see four plausible reasons, none of which we tested. (i) Instability is high everywhere because the models have almost no skill. When most of the output is noise, there is little room for volatility to make things worse. (ii) Small effects may exist below

our resolution: CI half-widths on regime differences range from 0.016 to 0.072. (iii) The test period (2021–2026) was calmer than the training years, so "volatile" days are relative to a high training median, and 7 of 55 folds had none. (iv) Volatile days have larger feature values, which may make the leading feature stand out more clearly. That would fit the two MLP results.

What practitioners should do. Report explanation stability across seeds and resamples, the explainer's own noise floor, and a no-skill control next to any feature-importance plot. Prefer deterministic explainers (TreeSHAP) or stabilised ones (S-LIME [5]) over default LIME, and interpret global rankings rather than single predictions.

8 Limitations

- **Next-day direction is close to random.** Accuracy is 0.507–0.512 against 0.502 for the control. Much of the instability we measure is the instability of explaining noise. Our conclusions apply to models of this quality and may not hold for a model with real skill.
- **Close-only data.** Coin Metrics Community data have no open/high/low, so range-based volatility and candle features are missing, and the regime uses close-to-close volatility only. `PriceUSD` is a cross-exchange reference rate, not an exchange's own close.
- **LTC instead of SOL.** SOL's volume history is missing in this dataset. The five coins move together, so clusters are not fully independent across coins, and our CIs may be somewhat optimistic.
- **Explained-sample size.** 10 points per (coin, fold), 550 in total (240 volatile), with 3 repeats for the stochastic explainers. CI half-widths on regime differences of 0.016–0.072 cannot rule out smaller effects.
- **Correlated features.** The MACD trio is exactly collinear, and the return windows overlap. Attribution methods can split credit among such features arbitrarily, which inflates top-k instability. A grouped analysis (`FEATURE_GROUPS` in `features.py`) was not run.
- **Fixed settings.** One configuration per model, one regime definition, one perturbation scheme, and top-k with k = 3 only. KernelSHAP used shap's default `l1_reg`, which zeroes 2 of 12 attributions per point.

9 Conclusion and Future Work

For next-day direction models on five cryptocurrencies, SHAP and LIME explanations fail basic trust checks. Methods disagree, LIME disagrees with itself, and the top driver changes with the random seed and with 5%-of-a-std input noise, even for TreeSHAP, whose own noise is zero. A no-skill control is just as stable. The models themselves barely beat a coin flip, and only in volatile regimes. On the key question the answer is a clear null: **explanation instability is not worse in volatile regimes**. Per-prediction feature-importance claims from such models should not be read as statements about the market, in any regime.

Future work. (1) Raise the explained sample and repeats; the full run takes about 5 minutes, so this is cheap. (2) Repeat the audit on *grouped* features to separate real instability from credit-splitting among correlated features. (3) Add OHLC or exchange-level data, SOL and more coins, and alternative regime definitions. (4) Apply the same audit to models with real skill (other horizons or targets) to see whether skill buys stability. (5) Compare stabilised explainers (S-LIME [5], interventional TreeSHAP) and test sensitivity to k.

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Note on the proposal's reference list. References [1]–[9] are the proposal's list. Krishna et al. [10], the core reading for the disagreement analysis (proposal Week 6), was missing and has been added. The proposal's timeline cited [5] (S-LIME) and [7] (crypto forecasting) in swapped weeks. Here S-LIME is cited for LIME's sampling instability (Sections 2, 6.2, 7, 9), and [7] for ML crypto forecasting (Sections 1, 2). Ribeiro et al. [4] is cited only as the paper that introduced LIME. [11]–[16] support specific methods used here.